Angelo Corsaro, Ph.D.

Google Scholar: scholar.google.com/citations?user=AngeloCorsaro | linkedin.com/in/corsaro

RESEARCH PROFILE

Dr. Angelo Corsaro is a world expert in Internet-scale distributed systems, with 25+ years of contributions spanning protocol design, real-time middleware, consensus and clock synchronization algorithms, and cloud-to-edge computing. His work has had both deep theoretical impact (100+ peer-reviewed publications, 1,478+ citations) and transformative industrial reach (10+ international standards co-authored, multiple protocols deployed in millions of mission-critical systems).

His research contributions include: **(i)** the invention of Zenoh, a protocol unifying pub/sub, query, and computation across the cloud-to-microcontroller continuum, now adopted in robotics (ROS 2), automotive (V2X), and IoT; **(ii)** pioneering work on Real-Time Java, including the first ahead-of-time compiled RTSJ implementation (jRate) and a constant-time algorithm for memory-reference safety verification; **(iii)** co-architecture of the OMG Data Distribution Service (DDS) standard; **(iv)** seminal contributions to fog computing, including direct collaboration with the Cisco team that coined the term, key contributions to the OpenFog Consortium reference architecture (IEEE 1934), and the creation of Eclipse fog05—the first fully decentralized fog computing infrastructure; and **(v)** novel distributed clock synchronization algorithms with the lowest error bounds published at the time (IEEE TPDS).

RESEARCH INTERESTS

- Distributed algorithms: consensus, clock synchronization, causal ordering, conflict-free replicated data types (CRDTs)
- Cloud-to-edge-to-microcontroller computing continuum: protocol design, data-centric architectures, zero-copy transport
- Real-time and safety-critical systems: deterministic scheduling, memory safety, formal verification
- Distributed storage and large-scale data transfer: alignment protocols, erasure coding, geo-distributed replication
- High-performance networking: zero-overhead abstractions, kernel bypass, RDMA-based middleware
- AI-native distributed infrastructure: federated learning, distributed inference, model-parallel communication

EDUCATION

Ph.D. in Computer Science

Washington University in St. Louis, Missouri, USA — *December 2004*

Dissertation: Design and performance of ahead-of-time compiled Real-Time Java middleware. Advisor: Prof. Douglas C. Schmidt. Research on jRate, memory-safety algorithms, and real-time distributed systems.

M.S. in Computer Science

Washington University in St. Louis, Missouri, USA — *January 2001*

Laurea in Computer Engineering (magna cum laude)

Università degli Studi di Catania, Italy — *July 1999*

SEMINAL RESEARCH CONTRIBUTIONS

Eclipse Zenoh: Unifying Communication, Storage, and Computation (2015–present)

Invented **Zenoh**, a protocol and middleware that unifies pub/sub, geo-distributed queries, and computation across the full cloud-to-microcontroller continuum. Zenoh achieves 5-byte wire overhead, sub-13µs latency, and 50 Gbps throughput, while running on devices with as little as 32 KB RAM. Selected by the ROS 2 Technical Steering Committee as the official DDS alternative middleware. Adopted by General Motors (uProtocol), Bosch, Volvo, Foxconn, and Volocopter. Recommended by ITU for Intelligent Transport Systems.

Key publications:

- Corsaro, A., Cominardi, L., Hecart, O., et al. (2023). "Zenoh: Unifying communication, storage and computation from the cloud to the microcontroller." 26th Euromicro Conference on Digital System Design (DSD), pp. 422–428.
- Baldoni, G., Loudet, J., Cominardi, L. & Corsaro, A. (2024). "Data-centric service-based architecture for edge-native applications." IEEE Communications Magazine, 62(4), pp. 32–38.
- Baldoni, G., Loudet, J., Cominardi, L. & Corsaro, A. (2021). "Facilitating distributed data-flow programming with Eclipse Zenoh." IEEE/ACM Symposium on Edge Computing.

Fog Computing: From "Fluid Computing" to IEEE Standard (2009–2020)

Pioneered the concept of distributing computation, storage, and networking to the edge of the network years before the term "fog computing" was coined—initially using the term ***Fluid Computing*** to capture the idea of computation flowing seamlessly between cloud and edge. Subsequently collaborated directly with the Cisco team that coined the term "fog computing," co-authoring the seminal IEEE Internet Computing article on applying fog computing to Barcelona's smart city infrastructure—a flagship initiative involving Cisco, the Barcelona City Council, the Barcelona Supercomputing Center, and PrismTech (where Corsaro served as CTO).

Served as a **key member of the OpenFog Consortium** (founded by Cisco, Intel, Microsoft, ARM, Dell, and Princeton), providing seminal contributions to the official definition and reference architecture of fog computing. This work was adopted as **IEEE Standard 1934-2018**, the first and foundational standard for fog computing worldwide.

Created **Eclipse fog05** (fogØ5, a hacker-style rendering of "fogOS")—a fully decentralized fog computing infrastructure that unifies compute, networking, and storage fabrics end-to-end from cloud to microcontroller. Unlike contemporaneous approaches that merely extended cloud paradigms to the edge, fog05 was designed from first principles for the extreme heterogeneity and resource constraints of fog environments: it supported seamless orchestration across x86 servers, ARM boards, and microcontrollers without requiring a centralized control plane. Published at IEEE CIoT 2018 and released as an Eclipse Foundation open-source project, fog05 anticipated by several years the architectural patterns now pursued by Kubernetes edge distributions, ETSI MEC, and similar efforts. fog05's architecture directly informed the design of Zenoh.

Key publications:

- Yannuzzi, M., van Lingen, F., Jain, A., Parellada, O.L., Flores, M.M., Carrera, D., … Corsaro, A. & Olivé, A. (2017). "A new era for cities with fog computing." IEEE Internet Computing, 21(2), 54–67.
- Baldoni, G., Corsaro, A., et al. (2018). "fogØ5: Unifying the computing, networking and storage fabrics end-to-end." IEEE Conference on Internet of Things (CIoT).
- Markakis, E.K., et al. incl. Corsaro, A. (2017). "EXEGESIS: Extreme edge resource harvesting for a virtualized fog environment." IEEE Communications Magazine, 55(7), 173–179.

OMG Data Distribution Service (DDS) Standard (2003–2015)

Founding Co-Chair of the OMG DDS Special Interest Group and major architect of the DDS family of standards (DDS, DDSI-RTPS, DDS Security, DDS RPC, DDS4CCM, ISO C++ DDS API). DDS is now the foundational real-time pub/sub standard deployed in millions of mission-critical systems worldwide: air traffic control, naval combat management, autonomous vehicles, medical devices, and industrial SCADA systems.

Distributed Clock Synchronization (2005–2009)

Invented a fully distributed internal clock synchronization algorithm based on coupling theory that achieved the lowest error bounds published at the time for dynamic large-scale systems. Unlike NTP-based hierarchical approaches, this algorithm operates without a master clock and tolerates dynamic topology changes. Published in IEEE Transactions on Parallel and Distributed Systems (2009) and deployed in Finmeccanica/SELEX-ES fault-tolerant systems.

Key publication:

- Corsaro, A., Baldoni, R., Querzoni, L., et al. (2009). "Coupling-based internal clock synchronization for large-scale dynamic distributed systems." IEEE TPDS, 21(5), pp. 607–619.

Novel Algorithms for Exotic Multi-Core Architectures (2005–2007)

As Technical Manager of the Finmeccanica Multi-Core Processor Research Group, led research on the IBM Cell Broadband Engine and other exotic architectures. Developed novel data-processing algorithms achieving 9x–40x performance improvements and 9x reduction in heat dissipation compared to conventional approaches. This work demonstrated that heterogeneous multi-core architectures could deliver supercomputer-class performance for embedded signal processing and data fusion applications.

Real-Time Java: jRate and Memory-Safety Algorithms (2001–2004)

Created **jRate**, the first open-source ahead-of-time compiled implementation of the Real-Time Specification for Java (RTSJ), built as an extension of GCJ. Developed within the DARPA PCES (Program Composition for Embedded Systems) program, jRate was the first RTSJ implementation capable of running on resource-constrained embedded platforms with deterministic timing guarantees. Boeing used jRate to prove the feasibility of Real-Time Java for flight-critical UAV avionics software—demonstrating for the first time that Java could meet hard real-time constraints for safety-critical aerospace applications, a result widely considered impractical at the time. This proof of feasibility led Boeing to subsequently adopt OVM (a commercially supported real-time JVM from Purdue) for production deployment. The jRate work was conducted in direct competition with leading research groups at MIT (Martin Rinard) and Purdue (Jan Vitek) under the PCES program.

Independently solved a fundamental theoretical problem in RTSJ memory safety: developed a **constant-time algorithm for verifying the safety of pointer assignments** in scoped memory regions. The RTSJ's single-parent rule requires runtime checks on every memory reference to prevent dangling pointers across scoped memory areas. The naïve approach is O(n) in scope nesting depth. Corsaro and Cytron's algorithm achieves O(1) verification through a novel encoding scheme — published at **LCTES 2003** (co-located with PLDI), this was the first solution to this problem and preceded all competing approaches.

Key publications:

- Corsaro, A. & Cytron, R.K. (2003). "Efficient memory-reference checks for real-time Java." ACM SIGPLAN Notices, 38(7), pp. 51–58. (LCTES '03, co-located with PLDI)
- Corsaro, A. & Schmidt, D.C. (2003). "The design and performance of real-time Java middleware." IEEE Transactions on Parallel and Distributed Systems, 14(11), pp. 1155–1167.
- Corsaro, A. & Schmidt, D.C. (2002). "The design and performance of the jRate real-time Java implementation." OTM Confederated International Conferences.

- Corsaro, A. & Schmidt, D.C. (2002). "Evaluating real-time Java features and performance for real-time embedded systems." Proc. 8th IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS), pp. 90–100.

PROFESSIONAL EXPERIENCE

CEO / CTO & Co-Founder — ZettaScale Technology, Paris

Jan 2022 – 2026

Co-founded company commercializing Eclipse Zenoh. Led the protocol from research prototype to production release (Zenoh 1.0, Oct 2024) within the Eclipse Foundation. Directed all R&D, technology strategy, and scientific publications. Team grew to include researchers and engineers across distributed systems, networking, and embedded computing. Selected to represent France at the Genius Minds 2024 initiative (Business France / Choose France) at the French Embassy in London. Spring50 finalist (Paris-Saclay SPRING). Continued publishing peer-reviewed research alongside executive responsibilities.

Chief Technology Officer — ADLINK Technologies Inc., Paris

2016 – 2022

Led the Advanced Technology Office. Established edge computing as strategic corporate pillar years before mainstream adoption. Directed technology scouting, R&D strategy, and standardization activities. Recognized as Power 200: Most Influential Data Economy Leader (2019).

Chief Technology Officer — PrismTech, Orsay

2009 – 2016

Transformed company from niche aerospace middleware vendor to IoT platform leader. Architected Vortex, the first commercial platform supporting unified cloud and fog computing (2009—2 years before the term "fog computing" was coined). Drove ~75% revenue growth. Gartner Cool Vendor (2014).

Product Strategy Manager — PrismTech, Orsay

2007 – 2009

Crafted and executed open-source strategy that drove massive user adoption and market expansion.

Software Technologies Scientist — SELEX-ES / Finmeccanica, Rome

Jan 2005 – Aug 2007

Strategic and Technological Planning Directorate. Technical Manager of Finmeccanica Multi-Core Processor Research Group. Chaired scientific committee for Finmeccanica Software Laboratory. Co-advised Ph.D. students at La Sapienza, Rome.

Chief Architect, R&D — SELEX-ES / Finmeccanica, Rome

Dec 2003 – Dec 2004

Designed enterprise middleware platform for mission-critical air traffic control, airborne, and naval combat management systems. Led international team of ~50 engineers.

TEACHING EXPERIENCE

Lecturer — ESIEE Paris (member of the Conférence des Grandes Écoles Françaises)

2016 – 2022

Courses taught: Internet of Things (IoT), Functional Programming. Bridged academic research with industrial practice, preparing engineering students for emerging distributed computing paradigms.

Ph.D. Co-Advisor

Co-advised doctoral students at Università La Sapienza (Rome) and Washington University in St. Louis on topics in distributed systems, real-time middleware, and fault-tolerant computing.

SELECTED PUBLICATIONS

Full list: 100+ peer-reviewed publications | 1,478+ citations (Google Scholar) | H-index reflecting sustained two-decade impact

Journal Articles (Peer-Reviewed)

- Corsaro, A. & Cytron, R.K. (2003). "Efficient memory-reference checks for real-time Java." ACM SIGPLAN Notices, 38(7), 51–58. [LCTES '03 / co-located with PLDI]
- Corsaro, A. & Schmidt, D.C. (2003). "The design and performance of real-time Java middleware." IEEE Trans. on Parallel and Distributed Systems, 14(11), 1155–1167.
- Corsaro, A., Baldoni, R., Querzoni, L., et al. (2009). "Coupling-based internal clock synchronization for large-scale dynamic distributed systems." IEEE TPDS, 21(5), 607–619.
- Baldoni, G., Loudet, J., Cominardi, L. & Corsaro, A. (2024). "Data-centric service-based architecture for edge-native applications." IEEE Communications Magazine, 62(4), 32–38.
- Corsaro, A., Yannuzzi, M., et al. (2017). "A new era for cities with fog computing." IEEE Internet Computing, 21(2), 54–67.
- Markakis, E.K., et al. incl. Corsaro, A. (2017). "EXEGESIS: Extreme edge resource harvesting for a virtualized fog environment." IEEE Communications Magazine, 55(7), 173–179.

Conference Papers (Peer-Reviewed)

- Corsaro, A., Cominardi, L., Hecart, O., et al. (2023). "Zenoh: Unifying communication, storage and computation from the cloud to the microcontroller." Euromicro DSD, 422–428.
- Corsaro, A. & Schmidt, D.C. (2002). "Evaluating real-time Java features and performance for real-time embedded systems." 8th IEEE RTAS, 90–100.
- Corsaro, A. & Schmidt, D.C. (2002). "The design and performance of the jRate real-time Java implementation." OTM Confederated International Conferences.
- Baldoni, G., Loudet, J., Cominardi, L. & Corsaro, A. (2021). "Facilitating distributed data-flow programming with Eclipse Zenoh." IEEE/ACM SEC.
- Krishna, A.S., Klefstad, R., Schmidt, D.C. & Corsaro, A. (2003). "Towards predictable real-time Java object request brokers." 9th IEEE RTAS, 237–248.
- Wang, N., Schmidt, D.C., van't Hag, H. & Corsaro, A. (2008). "Toward an adaptive data distribution service for dynamic large-scale NCOW systems." MILCOM, 1–7.

Book Chapters

- Corsaro, A. & Schmidt, D.C. (2012). "The Data Distribution Service — the communication middleware fabric for scalable and extensible systems-of-systems." System of Systems, Ch. 19. IntechOpen.
- Corsaro, A. et al. (2006). "Quality of service in publish/subscribe middleware." Global Data Management, 1–22. IOS Press.
- Krishna, A.S., Schmidt, D.C., Klefstad, R. & Corsaro, A. (2004). "Real-time CORBA middleware." Middleware for Communications, 413–438. Wiley.

INTERNATIONAL STANDARDS & OPEN-SOURCE LEADERSHIP

Standards Co-Authored (~10+)

- OMG Data Distribution Service (DDS) v1.0–1.4 — Founding Co-Chair (2004–2015)
- OMG DDSI-RTPS (interoperability wire protocol) v2.x — Contributor
- OMG DDS Security, DDS RPC, DDS4CCM — Co-author

- ISO C++ DDS API (SimD) — Primary author; adopted as OMG standard
- 5+ additional OMG specifications in real-time and embedded systems
- OpenFog Consortium Reference Architecture — Key contributor; adopted as IEEE Standard 1934-2018

Open-Source Projects (Creator/Lead)

- Eclipse Zenoh — Inventor and project lead; Eclipse Foundation Board of Directors
- Eclipse Cyclone DDS — Co-lead
- Eclipse fog05 (fogØ5) — Creator and project lead; first fully decentralized fog computing infrastructure
- jRate — Creator; first open-source ahead-of-time compiled RTSJ implementation
- SimD — Sole author; ISO C++ DDS API (adopted as OMG standard)
- Dada — Scala library for distributed algorithms

INVITED TALKS & KEYNOTES

400+ invited talks and tutorials globally. Selected recent keynotes:

- DSN '25 (IEEE/IFIP International Conference on Dependable Systems and Networks)
- EUROMICRO-DSD '25 (Euromicro Conference on Digital System Design)
- GoSim '25 (Paris)
- OOPSLA Tutorial: OMG Data Distribution Service (DDS)
- W3C Workshop: DDS for the Web of Things (Berlin, 2014)

RECOGNITION & AWARDS

- Technology CEO of the Year (2024)
- Best Cloud-to-Device Continuum Platform Award (2024)
- Top 10 V2X Startup to Watch (2023, 2024)
- Genius Minds 2024 — Selected to represent France at the French Embassy in London (Business France / Choose France initiative)
- Spring50 Finalist — Paris-Saclay SPRING, selected among the 50 most innovative deep tech startups of the Paris-Saclay ecosystem (ZettaScale)
- Power 200: World's Most Influential Data Economy Leaders (2019)
- Compass Intelligence Award: Best M2M & IoT Company (2018)
- Top 50 Most Influential Executives in Edge Computing (2017–2018)
- Gartner Cool Vendor (2014)
- IEEE Senior Member | ACM Member
- Eclipse Foundation Board of Directors

LANGUAGES

Natural: Italian (native), English (fluent), French (fluent)

Programming: Rust (expert), C/C++ (25+ years), OCaml (expert), Java (expert — real-time Java pioneer), Scala (expert), Python, JavaScript

RESEARCH VISION STATEMENT

The convergence of ubiquitous computing, artificial intelligence, and the regulatory imperative for digital sovereignty is creating an urgent need for a new generation of distributed systems — systems that are formally verifiable, that operate seamlessly from cloud data centers to microcontrollers, and that can enforce data residency and provenance at the protocol level. My research program addresses this challenge through the co-design of distributed algorithms, protocol architectures, and systems software. I am particularly interested in bringing rigorous theoretical foundations — from consensus theory, type systems, and formal verification — to bear on practical systems that must operate at Internet scale with real-time guarantees. I believe that academic-industry collaboration is essential to this mission, and I am eager to contribute both my research depth and my two decades of experience translating theory into deployed systems.

Complete publication list and references available upon request.